DiaTool-DPO:

Multi-Turn DPO for Tool-Augmented LLMs

Enhancing TA-LLM dialogue flow via direct preference optimization with automatic paired trajectory construction.

SFT Alone Fails to Teach Slot-Filling and Tool Rejection

The Core Problem

TA-LLMs trained only with Supervised Fine-Tuning (SFT) struggle with:

- **Incomplete queries (missing arguments)**
- **Out-of-scope requests**

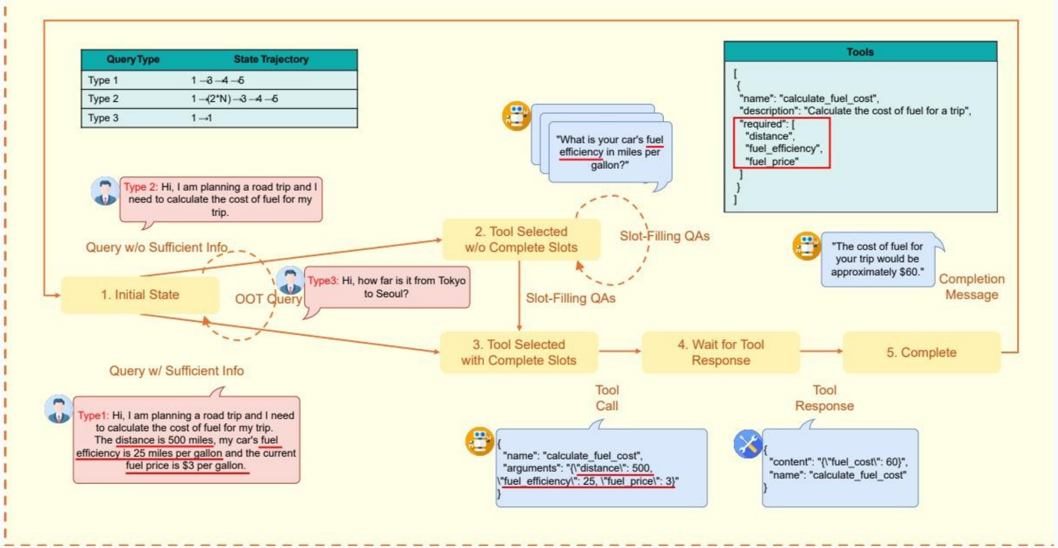
Three Failure Modes

1. Failing to ask follow-up questions

2. Making hallucinated tool invocations

3. Failing to reject inappropriate tool calls

Modeling TA-LLM Dialogue as a 5-State Markov Decision Process



5 Internal States

1. Initial State
No history. Reject out-of-scope.

2. Tool Selected w/o Complete Slots
Slot-filling QA occurs.

3. Tool Selected w/ Complete Slots
Tool selection & arguments determined.

4. Wait for Tool Response
Awaiting results.

5. Complete

Three Query Types Define Distinct State Trajectories

Query Type	State Trajectory	Description & Example
Type 1	1 → 3 → 4 → 5	Query contains all required arguments. No slot-filling needed.\nExample: 'What is the weather in New York today?'
Type 2	1 → (2×N) → 3 → 4 → 5	Query lacks some arguments. Slot-filling QA repeats N times until complete.\nExample: 'Book me a flight to London.' (Missing date & origin)
Type 3	1 → 1	Requested functionality is unavailable. Tool call must be rejected.\nExample: 'Who won the Super Bowl in 2024?' (No suitable tool available)

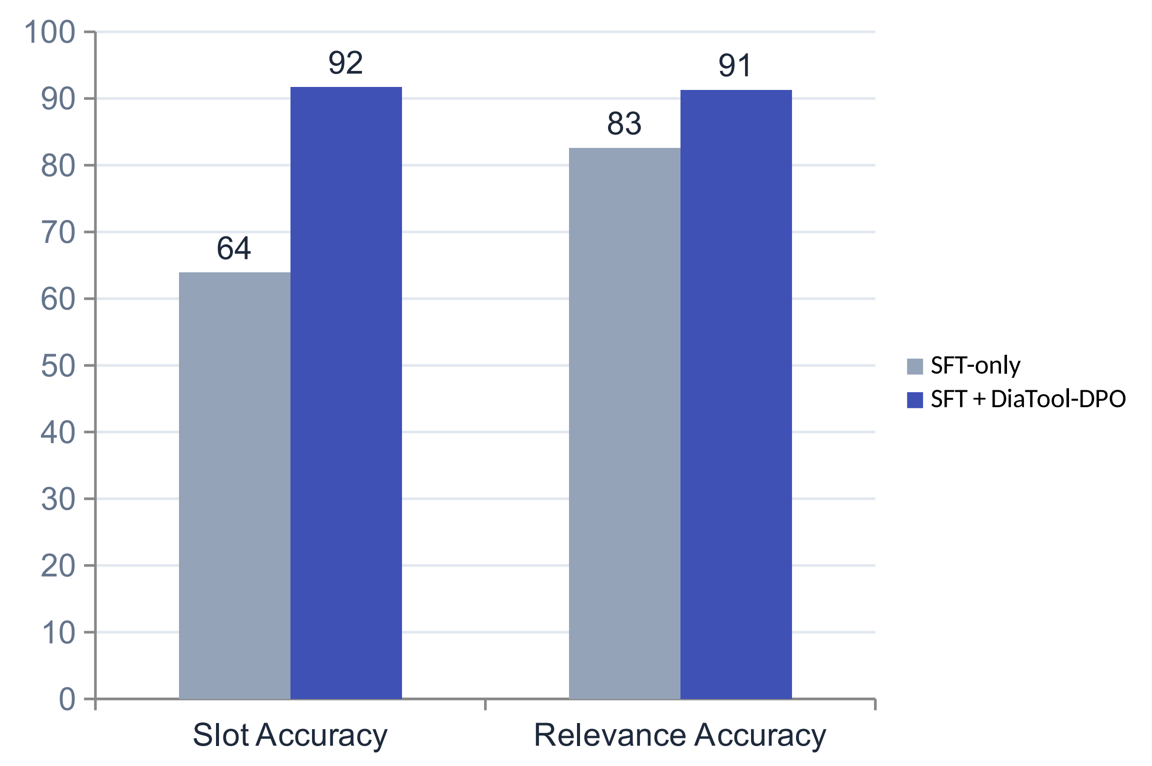
Automatic Paired Trajectory Construction Without Human Labels

Query	Chosen Traj.	Rejected Traj.	Learning Lesson	Count	Difficulty
Type 1	1→3→4→5	1→2→3→4→5	Prevent redundant slot-filling	2,089	Easy
Type 1	1→3→4→5	1→(2*N)→3→4→5	Prevent redundant slot-filling	562	Hard
Type 1	1→3→4→5	1→(2*M)→3→4→5	Prevent redundant slot-filling	2,530	Hard
Type 1	1→3→4→5	1→1	Tool call accept	2,090/562	Easy, Hard
Type 2	1→2→3→4→5	1→3→4→5	Prevent slot hallucination	2,089	Easy
Type 2	1→(2*N)→3→4→5	1→3→4→5	Prevent slot hallucination	562	Hard
Type 2	1→(2*N)→3→4→5	1→(2*M)→3→4→5	Prevent slot hallucination	2,530	Hard
Type 2	—	1→1	Tool call accept	2,089/562	Easy, Hard
Type 3	1→1	1→3→4	Tool call reject	567	Hard
Type 3	1→1	1→(2*N)→3→4	Tool call reject	562	Hard

SFT + DiaTool-DPO Achieves Best Macro Average Across All Models

Model	Method	Call	Completion	Slot	Relevance	Micro Avg.	Macro Avg.
Prop.-8B	DiaTool-DPO-only	0.314	0.700	0.833	0.609	0.575	0.614
Prop.-8B	SFT-only	0.900	0.916	0.694	0.913	0.870	0.856
Prop.-8B	**SFT + DiaTool-DPO**	0.886	0.929	0.833	0.826	0.884	**0.868**
Prop.-3.1B	DiaTool-DPO-only	0.357	0.551	0.528	0.391	0.455	0.457
Prop.-3.1B	SFT-only	0.771	0.817	0.750	0.826	0.790	0.791
Prop.-3.1B	**SFT + DiaTool-DPO**	0.743	0.871	0.833	0.826	0.765	**0.818**
LLaMA-3-8B	DiaTool-DPO-only	0.029	0.449	0.056	0.261	0.205	0.199
LLaMA-3-8B	SFT-only	0.843	0.957	0.639	0.826	0.844	0.816
LLaMA-3-8B	**SFT + DiaTool-DPO**	0.857	0.929	0.917	0.913	0.905	**0.904**

LLaMA-3-8B: +43.5% Slot Accuracy and +10.5% Relevance



Massive Improvements

+43.5%

in Slot Filling

+10.5%

in Tool Rejection

Key Takeaways

RL-Based Dialogue Control for Production TA-LLMs

MDP Formulation Enables Systematic Pairing

Defining 5 internal states and 3 query types structures the dialogue control space.

No Human Labeling Required

Automatic construction of chosen/rejected trajectories vastly reduces dataset cost.

DPO Complements SFT

DPO alone performs poorly, but SFT + DiaTool-DPO achieves the best macro average.

Language-Agnostic & Broadly Applicable